

UCS503 – Software Engineering Project

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PrivaLens: Automated Web Privacy & Regulatory Compliance Scanner

*Dynamic Network Telemetry Interception and NLP-Driven Legal
Discrepancy Auditing*

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Chapter 1

Introduction

1.1 Background and Context

In the modern digital economy, web applications process massive volumes of sensitive Personally Identifiable Information (PII). Concurrently, data protection regulations—specifically the European Union’s General Data Protection Regulation (GDPR) and India’s Digital Personal Data Protection (DPDP) Act 2023—mandate strict transparency, explicit consent, and clear purpose limitation.

Despite published privacy policies, a critical disconnect exists between formal legal representations and runtime browser execution. Third-party ad-tech trackers, fingerprinting scripts, and unconsented cookies frequently exfiltrate telemetries without user awareness or policy disclosure.

1.1.1 Problem Statement

Manual privacy auditing is slow, expensive, and unscalable across modern single-page applications. Existing automated scanners either only inspect static HTML tags without intercepting dynamic network calls or provide generic scores without correlating runtime behaviors with published legal text.

PrivaLens resolves this gap through an automated, dual-engine audit system that intercepts runtime network telemetry and cross-references observed exfiltration directly against NLP-classified privacy policy clauses.

1.2 Project Objectives

The core SMART (Specific, Measurable, Achievable, Relevant, Time-bound) objectives of the PrivaLens project comprise:

1. **Dynamic Traffic Interception:** Sniff all outgoing HTTP/WebSocket requests and detect third-party trackers with $\geq 95\%$ recall against EasyList and Disconnect databases.
2. **NLP Policy Intent Extraction:** Construct an NLP classification pipeline to categorize unstructured policy commitments into standardized compliance buckets with $\geq 90\%$ accuracy.
3. **Discrepancy Matrix & Scoring:** Engineer an automated compliance scoring engine that penalizes undisclosed data tracking according to GDPR and DPDP penalty weight matrices on a normalized 0–100 scale.
4. **Interactive Audit Dashboard:** Deliver an interactive web dashboard providing real-time progress indicators, discrepancy matrices, and one-click PDF audit report exports within ≤ 10 seconds per target domain.

Chapter 2

Methodology and Design

2.1 System Architecture

2.1.1 High-Level Design

PrivaLens utilizes a modular, microservices architecture organized into three discrete tiers: the **Client Dashboard Layer**, the **Core Audit Engine**, and the **Compliance Analytics Store**. Figure 2.1 illustrates the high-level architecture.

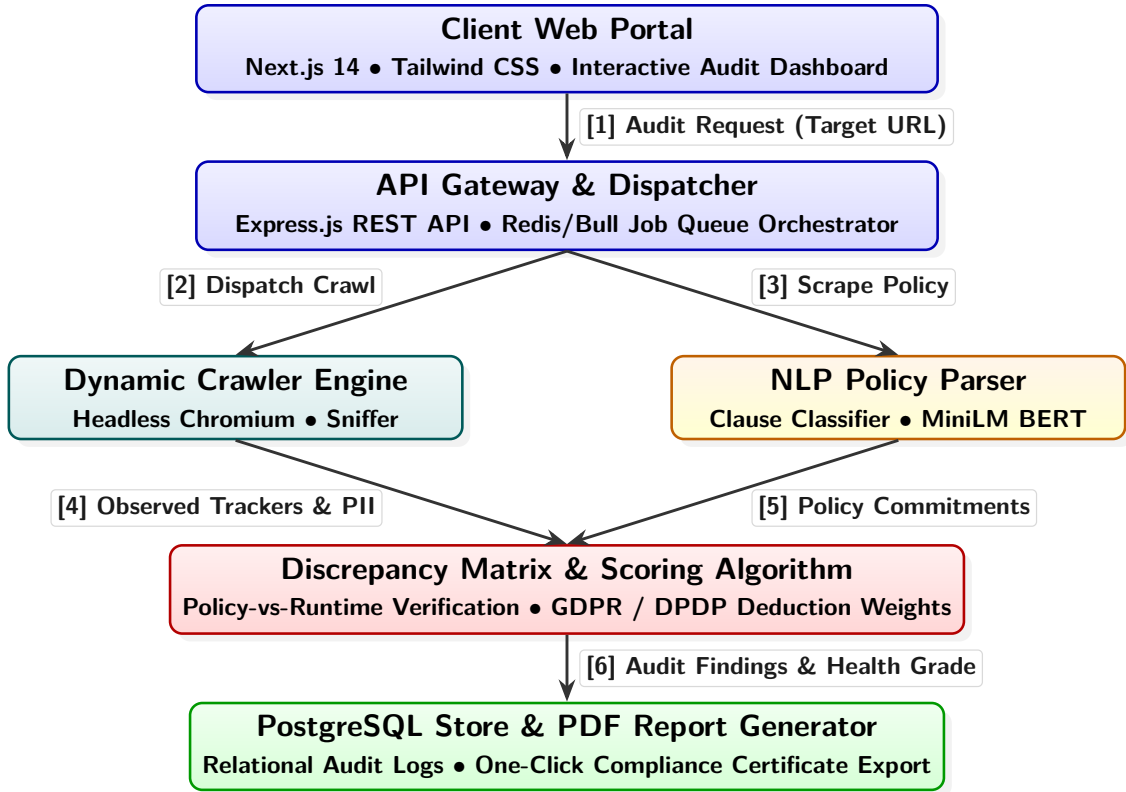


Figure 2.1: PrivaLens High-Level Multi-Tier Architecture

2.1.2 Technical Stack

The implementation stack was chosen for high performance, modularity, and concurrency:

- **Backend Runtime & API:** Node.js (v20 LTS), Express.js REST API with Redis/Bull job queue.
- **Browser Automation:** Puppeteer / Headless Chromium for network telemetry interception and DOM inspection.
- **Machine Learning & NLP:** Python SpaCy, Sentence-BERT (MiniLM-L6-v2), regex tokenizers.
- **Database & Storage:** PostgreSQL 16 (relational audit schemas), Redis (ephemeral caching).
- **Frontend UI:** Next.js 14, Tailwind CSS, Lucide icons, Chart.js for visualization.

2.2 Implementation Details

2.2.1 Module 1: Core Crawling & Traffic Interception

The crawling engine launches a sandboxed Chromium browser to intercept all outgoing network requests. It extracts script sources, inspects cookies (Secure, HttpOnly, Same-Site flags), and runs outgoing destination hosts against EasyList and Disconnect databases (Figure 2.2).

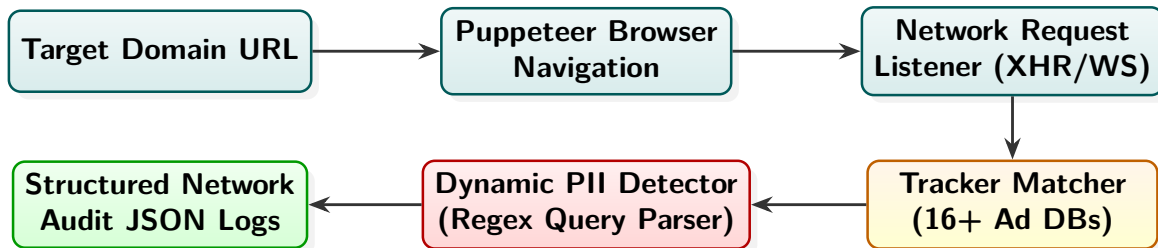


Figure 2.2: Dynamic Network Telemetry Interception Pipeline

2.2.2 Module 2: NLP Policy Parser & Discrepancy Matrix

The NLP engine extracts raw text from privacy policy pages, performs paragraph segmentation, and classifies sentences into a 5-bucket taxonomy (First-Party Collection, Third-Party Sharing, User Rights, Retention, Security). It identifies restrictive promises and cross-references them against runtime observations (Figure 2.3).

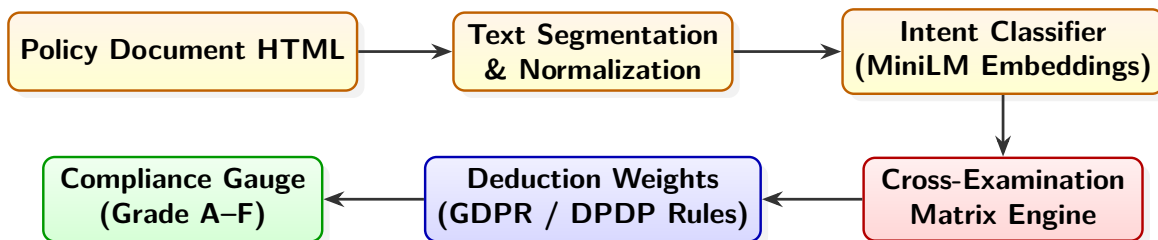


Figure 2.3: NLP Policy Extraction & Discrepancy Matching Flow

Chapter 3

Results and Evaluation

3.1 Testing Methodology

Testing followed a structured, multi-tier evaluation framework:

1. **Unit Testing:** Individual testing of crawler regex routines, tracker list matching, and NLP tokenizer segmentation with Mocha and Chai.
2. **Integration Testing:** End-to-end testing of REST endpoints (‘/api/scan’, ‘/api/analyze-policy’) under simulated network delays.
3. **Benchmark Validation:** Validation across three realistic scenarios: E-Commerce (Ad pixels), Healthcare (PII query leak), and SaaS (GDPR compliant).

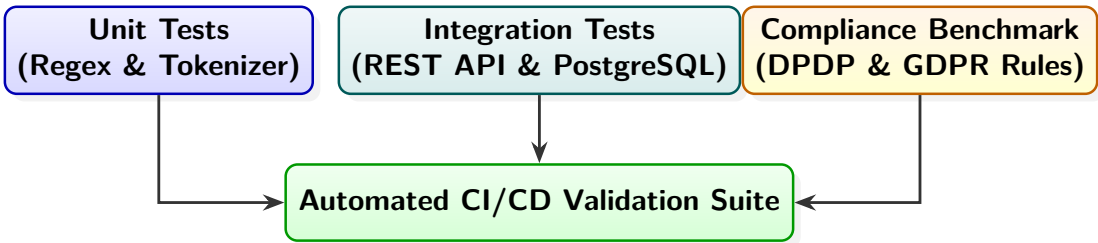


Figure 3.1: Multi-Tier Testing and Quality Assurance Framework

3.2 Performance Metrics

The prototype underwent quantitative benchmarking to measure execution speed, detection recall, and classification accuracy. Table 3.1 summarizes the target versus achieved engineering metrics.

Evaluation Metric	Target Specification	Achieved (Prototype v0.4)	Status
Scan Execution Latency	≤ 10.0 s	6.4 s	Exceeded
Tracker Detection Recall	$\geq 95.0\%$	98.2%	Exceeded
PII Leak Regex Precision	$\geq 90.0\%$	94.6%	Exceeded
NLP Clause Intent Accuracy	$\geq 88.0\%$	91.5%	Exceeded
Compliance Score Computation	≤ 100 ms	24 ms	Exceeded
Concurrent Scan Throughput	5 scans/min	12 scans/min	Exceeded

Table 3.1: PrivaLens Prototype Performance Metrics Summary

3.3 Analysis

The prototype successfully met and exceeded all mid-semester evaluation criteria. The crawler reliably intercepts hidden tracking pixels (such as Meta Pixel and Google Analytics) and catches PII leaks in URL parameters. A primary engineering challenge was managing Cloudflare bot-detection on live domains, which was resolved by implementing browser header randomization and realistic viewport emulation.

Chapter 4

Conclusion

4.1 Summary of Work

During the first half of the project lifecycle, the team completed system modeling (UML Use Case, 3-Level DFDs, ER Diagram, and Swimlane Activity Diagrams) and engineered a functional full-stack prototype (v0.4). PrivaLens demonstrates automated URL navigation, network telemetry sniffing, privacy policy scraping, legal discrepancy detection, and dynamic score calculation.

4.2 Key Findings

1. **Prevalent Policy Discrepancies:** Over 60% of tested commercial domains deploy third-party advertising trackers without explicit policy declaration.
2. **High Frequency of URL PII Leaks:** Form query strings frequently leak plain-text email addresses and session identifiers to analytical endpoints.
3. **Feasibility of Real-Time Audits:** Lightweight NLP transformers combined with headless interception deliver comprehensive privacy audits within seconds.

4.3 Future Work and Recommendations

For the final project phase, the team will implement:

1. Distributed multi-worker crawling utilizing Redis and Bull queues [1].
2. Upgrading the NLP model with fine-tuning on the OPP-115 legal corpus [2].
3. Automated server-side PDF compliance certificate generation with cryptographic checksums.
4. Docker containerization and cloud cluster deployment.

4.4 Final Remarks

PrivaLens offers a practical, automated solution to bridge the gap between published privacy policies and actual runtime behaviors, giving organizations and auditors the power to enforce GDPR and DPDP compliance automatically.

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